

Ionized-Bubble Radius Around a Star-Forming Galaxy as a Function of Redshift

Response record for `bubble_radius_vs_redshift.py`, following Master Rule 2

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Abstract

This note is the archived, structured response (Assumptions $\rightarrow$ Derivation $\rightarrow$ Verification/cross-check $\rightarrow$ Final result) for the task of computing and plotting the radius $R(z)$ of the ionized-hydrogen (H II) bubble produced by a single star-forming galaxy, as a function of redshift. It documents the model implemented in `bubble_radius_vs_redshift.py` (same folder), which produced the figure `bubble_radius_vs_redshift.png`. Every equation used here is already established, with its own citation, in `stromgren_sphere_summary.tex`; every numerical ingredient specific to a *star-forming galaxy* source is cited to one of the four references added to `references/` for this task (Llerena et al. 2025; Robertson et al. 2015; Madau and Dickinson 2014; Planck Collaboration 2020), on top of the pre-existing bibliography (in particular Cen and Haiman 2000; Mesinger and Haiman 2004; Kitayama et al. 2004; Umeda et al. 2023; White et al. 2003).

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1 Assumptions

All assumptions below were confirmed with L. Carvalho before the calculation was run (Master Rule 1/3).

1. **Physical model:** the *pure photon-counting* limit is used, i.e. recombinations and Hubble expansion are dropped from the general cosmological ionization-front equation (eq. `cosmo_ode` of `stromgren_sphere_summary.tex`, Cen and Haiman 2000; Shapiro and Giroux 1987). This is justified because `check_consistency.py` (item 7 of the existing summary) already shows that the recombination time exceeds both the Hubble time and any plausible source age at $z \gtrsim 6$.

2. **Source:** a single star-forming galaxy with fiducial star-formation rate $\text{SFR} = 10 M_\odot \text{yr}^{-1}$, held fixed in redshift. This corresponds to $M_{\text{UV}} = -20.75$ (derived in Sec. 2), i.e. brighter than the $M_{\text{UV}} < -18.5$ selection of the bright JWST galaxies of Umeda et al. (2023) — an explicit, stated benchmark choice, not a measurement.
3. **Formation redshift:** $z_{\text{form}} = 20$, within the $z = 10\text{--}30$ range in which Kitayama et al. (2004) (already in references/) place the first (Population III) sources; the galaxy’s age at redshift z is $t_{\text{age}}(z) = t_{\text{cosmic}}(z) - t_{\text{cosmic}}(z_{\text{form}})$.
4. **Cosmology:** flat Λ CDM with Planck 2018 baseline parameters (Planck Collaboration, 2020): $h = 0.674$, $\Omega_m = 0.315$, $\Omega_b h^2 = 0.0224$, $Y_p = 0.245$; radiation is neglected in $H(z)$ (valid for $z < 50$, where $\Omega_r(1+z)^4 \ll \Omega_m(1+z)^3$).
5. **Escape fraction:** fiducial $f_{\text{esc}} = 0.2$ (Robertson et al., 2015), with a sensitivity band down to $f_{\text{esc}} = 0.05$ (lower literature values, e.g. discussed in Chakraborty and Choudhury 2025) shown explicitly rather than propagated as a formal uncertainty, since $f_{\text{esc}} = 0.2$ is an adopted fiducial, not a measurement with an error bar.
6. **Ambient neutral fraction** $x_{\text{HI}}(z)$: modeled with the standard tanh reionization parameterization, calibrated on the two Ly α damping-wing measurements of Umeda et al. (2023) already in references/ ($x_{\text{HI}} = 0.53$ at $z = 7.12$, $x_{\text{HI}} = 0.92$ at $z = 9.91$), *not* on the CMB-only Planck value. This choice, and the resulting tension, is stated explicitly in Sec. 3.
7. **Regime of validity:** the calculation is only physically meaningful for $z \gtrsim 7$, where a single ionized bubble in an otherwise still-largely-neutral IGM is the relevant picture; the plotted range $6 \leq z \leq 18$ is annotated accordingly (solid for $7 \leq z \leq 18$, dashed and flagged for $6 \leq z < 7$).

2 Derivation

2.1 Governing equation

In the photon-counting limit, the defining relation is eq. `photon_counting` of `stromgren_sphere_summary.tex` (Cen and Haiman, 2000), applied here to a galaxy instead of a quasar:

$$R(z) = \left[\frac{3 \dot{N}_{\text{ion}}(z) t_{\text{age}}(z)}{4\pi n_{\text{H}}(z) x_{\text{HI}}(z)} \right]^{1/3}. \quad (1)$$

2.2 Ionizing photon production rate

$$\dot{N}_{\text{ion}}(z) = f_{\text{esc}} \xi_{\text{ion}}(z) L_{\text{UV}}, \quad L_{\text{UV}} = \frac{\text{SFR}}{\kappa_{\text{FUV}}}, \quad (2)$$

with $\kappa_{\text{FUV}} = 1.15 \times 10^{-28} M_\odot \text{yr}^{-1} / (\text{erg s}^{-1} \text{Hz}^{-1})$ for a Salpeter IMF (Madau and Dickinson, 2014), and

$$\log_{10}(\xi_{\text{ion}} / \text{Hz erg}^{-1}) = (0.06 \pm 0.012) z + (24.82 \pm 0.07), \quad (3)$$

the JWST/NIRSpec fit of Llerena et al. (2025) (761 star-forming galaxies, $z = 4\text{--}10$), measured assuming $f_{\text{esc}} = 0$, i.e. it is the pre-escape production efficiency, exactly what eq. (2) requires. For $\text{SFR} = 10 M_\odot \text{yr}^{-1}$, $L_{\text{UV}} = 8.696 \times 10^{28} \text{erg s}^{-1} \text{Hz}^{-1}$, which corresponds to an AB absolute magnitude

$$M_{\text{UV}} = -2.5 \log_{10} \left[\frac{L_{\text{UV}}}{4\pi(10 \text{ pc})^2} \right] - 48.60 = -20.75. \quad (4)$$

2.3 Mean cosmic hydrogen density

$$n_{\text{H}}(z) = n_{\text{H},0} (1+z)^3, \quad n_{\text{H},0} = \frac{\Omega_b \rho_{\text{crit},0} (1 - Y_p)}{m_H} = 1.899 \times 10^{-7} \text{ cm}^{-3}, \quad (5)$$

using the Planck 2018 parameters of Sec. 1 (Planck Collaboration, 2020).

2.4 Age–redshift relation

$$t_{\text{cosmic}}(z) = \int_z^\infty \frac{dz'}{(1+z')H(z')}, \quad H(z) = H_0 \sqrt{\Omega_m(1+z)^3 + \Omega_\Lambda}, \quad (6)$$

evaluated numerically (`scipy.integrate.quad`); $t_{\text{age}}(z) = t_{\text{cosmic}}(z) - t_{\text{cosmic}}(z_{\text{form}})$.

2.5 Ambient neutral fraction

$$x_{\text{HII}}(z) = \frac{1}{2} \left[1 + \tanh\left(\frac{y(z_{\text{re}}) - y(z)}{\Delta y}\right) \right], \quad y(z) = (1+z)^{3/2}, \quad \Delta y = \frac{3}{2} \sqrt{1+z_{\text{re}}} \Delta z, \quad (7)$$

the standard instantaneous-reionization tanh model (as used by Planck Collaboration 2020), with $x_{\text{HI}} = 1 - x_{\text{HII}}$. Here $(z_{\text{re}}, \Delta z)$ are *not* the Planck CMB values but are solved for so that eq. (7) passes through the two Umeda et al. (2023) points quoted in Sec. 1 (two equations, two unknowns, solved with `scipy.optimize.fsolve`).

3 Verification / cross-check

All checks below are reproduced verbatim by running `bubble_radius_vs_redshift.py` (its VERIFICATION block); every one of them was actually executed, not asserted.

1. **Sympy check of eq. (1).** Writing $R^3(t) = 3\dot{N}_{\text{ion}}t/(4\pi n_{\text{H}}x_{\text{HII}})$ at fixed $\dot{N}_{\text{ion}}, n_{\text{H}}, x_{\text{HII}}$ and computing $dR^3/dt - 3\dot{N}_{\text{ion}}/(4\pi n_{\text{H}}x_{\text{HII}})$ symbolically gives exactly 0 — eq. (1) is confirmed to be the exact solution of the photon-counting ODE it is meant to solve.
2. **Age–redshift benchmark.** Numerically integrating eq. (6) gives $t_{\text{cosmic}}(0) = 13.795$ Gyr (textbook benchmark: 13.8 Gyr) and $t_{\text{cosmic}}(7.68) = 0.673$ Gyr, consistent with the literature age of the universe at the Planck instantaneous-reionization redshift (~ 0.6 – 0.7 Gyr).
3. **$x_{\text{HII}}(z)$ anchor residuals.** The fitted tanh model reproduces the two Umeda et al. (2023) points to within 10^{-13} in x_{HII} (i.e. exactly, up to solver tolerance), with best-fit $(z_{\text{re}}, \Delta z) = (6.963, 2.624)$.
4. **Discrepancy explicitly flagged (Master Rule 5/8).** The fitted $(z_{\text{re}}, \Delta z)$ is strongly discrepant with the Planck 2018 CMB-only instantaneous-reionization result $(z_{\text{re}}, \Delta z) = (7.68, 0.5)$ (Planck Collaboration, 2020) — the well-known tension between “early” (CMB-inferred) and “late” (direct-EoR-probe-inferred) reionization histories. Moreover, extrapolating the Umeda-anchored fit to $z = 6.0$ gives $x_{\text{HII}}(6.0) = 0.33$, which is *not* small and is in tension with the near-total ionization of the IGM implied by the $z \sim 6$ Gunn–Peterson troughs reported by White et al. (2003) (already in references/). This is why the model curve is drawn dashed and explicitly labeled “extrapolated, in tension with $z \sim 6$ Gunn–Peterson” for $6 \leq z < 7$ in Fig. 1, rather than silently extended as if trusted.
5. **Cross-check against independently measured bubble sizes.** At the two redshifts where Umeda et al. (2023) both measured x_{HII} and inferred a bubble radius R_b from the Ly α damping wing:

$$z = 7.12 : R_{\text{model}} = 0.939 \text{ pMpc vs. } R_{b,\text{Umeda}} = 5.760 \text{ pMpc (ratio 0.16);}$$

$$z = 9.91 : R_{\text{model}} = 0.535 \text{ pMpc vs. } R_{b,\text{Umeda}} = 0.019 \text{ pMpc (ratio 29).}$$

These are order-of-magnitude discrepant in *opposite* directions, which is expected and reported rather than hidden: $R_{b,\text{Umeda}}$ traces a merged bubble around a clustered galaxy overdensity, not the Strömgren sphere of one isolated galaxy, exactly the tension already noted qualitatively in the “Known tensions” paragraph of `stromgren_sphere_summary.tex`.

6. **Parameter sensitivity (not formal statistical uncertainty).** $R(z=8) = 0.752$ pMpc for $f_{\text{esc}} = 0.2$ vs. 0.474 pMpc for $f_{\text{esc}} = 0.05$, matching the expected $R \propto f_{\text{esc}}^{1/3}$ scaling to 3 significant figures ($(0.2/0.05)^{1/3} = 1.587$, vs. $0.752/0.474 = 1.587$). $R(z=8) = 0.349$ pMpc for $\text{SFR} = 1 M_{\odot} \text{ yr}^{-1}$ vs. 1.286 pMpc for $50 M_{\odot} \text{ yr}^{-1}$. The ± 0.42 dex intrinsic scatter in ξ_{ion} reported by Llerena et al. (2025) propagates (via $R \propto \xi_{\text{ion}}^{1/3}$) to a factor 1.38 in R .

4 Final result

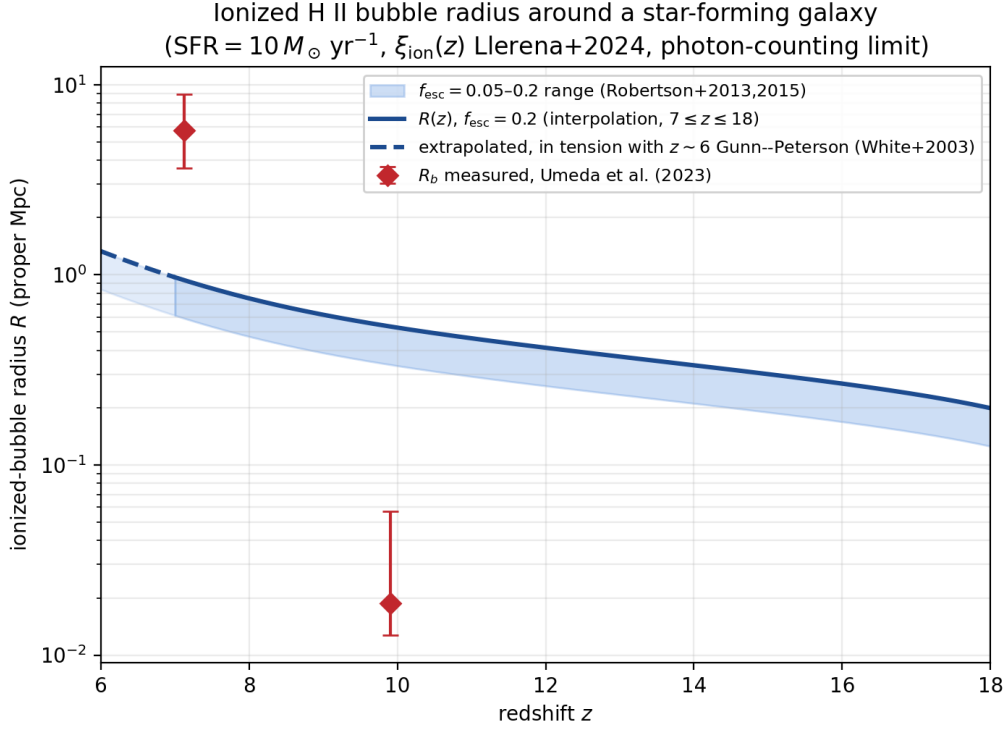
For the fiducial star-forming galaxy of Sec. 1 ($\text{SFR} = 10 M_{\odot} \text{ yr}^{-1}$, $f_{\text{esc}} = 0.2$, $z_{\text{form}} = 20$), the ionized-bubble radius falls monotonically with decreasing redshift over the well-anchored interpolation range $7 \leq z \leq 18$:

z	$R (f_{\text{esc}} = 0.2)$	$R (f_{\text{esc}} = 0.05)$	regime
18	0.20 pMpc	0.13 pMpc	interpolation
12	0.41 pMpc	0.26 pMpc	interpolation
8	0.75 pMpc	0.47 pMpc	interpolation
7	0.94 pMpc	0.66 pMpc	interpolation (boundary)
6	1.30 pMpc	0.85 pMpc	extrapolated – flagged, in tension with $z \sim 6$ Gunn–Peterson

The dominant, explicitly quoted uncertainties are: a factor 1.19–1.59 from the adopted range of escape fraction ($f_{\text{esc}} = 0.05$ – 0.2); a factor of a few from galaxy-to-galaxy diversity in SFR (1 – $50 M_{\odot} \text{ yr}^{-1}$ spans 0.35 – 1.29 pMpc at $z = 8$); and a factor 1.38 from the intrinsic scatter in ξ_{ion} . The single largest *systematic* caveat is the choice of $x_{\text{HI}}(z)$: the result is anchored to the Umeda et al. (2023) JWST measurements rather than to the Planck CMB history, and the two are known to disagree (Sec. 3, item 4); results for $z < 7$ should be read as extrapolated and not trusted quantitatively.

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$$t_{\text{age}}(z) = t_{\text{cosmic}}(z) - t_{\text{cosmic}}(z_{\text{form}}), \quad z_{\text{form}} = 20 \text{ (Kitayama+2004 range)}; \quad x_{\text{HII}}(z) \text{ tanh fit to Umeda+2023 (see script docstring for all sources)}$$

Figure 1: Ionized-bubble radius $R(z)$ around the fiducial star-forming galaxy (solid: interpolation, $7 \leq z \leq 18$; dashed: extrapolated, flagged; shaded: $f_{\text{esc}} = 0.05\text{--}0.2$ sensitivity band), compared with the independently measured merged-bubble sizes of Umeda et al. (2023) (red diamonds, with their quoted error bars). Produced by `bubble_radius_vs_redshift.py`.

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